

Chapter 65 — Adolescent and Transitional Urology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **Start transition ~age 12 (AAP)**; transfer 18–21; complete 18–23; latest **26**
2. **Paediatric team initiates**; Three Ps: Preparation, Plan, **named** Provider
3. **Transfer ≠ transition**; outcome metric = **stability** + retention in follow-up
4. **No transition during crisis**; exception **pregnancy** → co-management
5. **Ileal augment**: annual **B₁₂** from 5–10 y; bicarbonate/chloride
6. **After cystoplasty**: urine hCG 57–75 % false + → **serum hCG**
7. **Cancer**: ileal ~1.2 %, gastric **2.8–10.3 %**; cystoscopy **only if symptoms/filling defect**
8. **Perforation** 0.8–13 %; **mortality ≤ 25 % if delayed**
9. **SBA** (Spina Bifida Association): annual urinalysis, creatinine, renal and bladder ultrasound (RBUS); **cystatin C** if low muscle mass
10. **Bladder exstrophy-epispadias complex (BEEC) / cloaca / urinary reconstruction** → **planned caesarean**
11. **Bladder re-evaluation at transplant listing is mandatory**; first graft **left pelvis**
12. **SB**: normal IQ; ↓ **working memory + executive function**

WHAT CONSTITUTES ADOLESCENT UROLOGY?

- **Adult reconstructive urologist**: operatively capable; **underequipped for whole patient**
- **Aim**: seamless lifelong care; adolescent urology = **bridge**

DEFINITION OF TRANSITION

- **Got Transition**: child/family-centred → **adult/patient-centred** care
- **Three parts**: planning → transfer → integration

WHY IS TRANSITION NEEDED?

- **AHRQ** (Agency for Healthcare Research and Quality): **"no significant difference"**, but metrics inadequate
- **Avoid "normal"**: patient = **own baseline**

WHO NEEDS TRANSITION?

- **1. Complex congenital** (exstrophy, posterior urethral valves, complex hypospadias, SB) → **lifelong**
- **2. Quiescent** (vesicoureteric reflux, VUR) → **may be discharged**, open return
- **3. Cured, treatment sequelae** (pelvic rhabdomyosarcoma) → late-effects surveillance
- **4. Beyond paediatric scope** → fertility, pregnancy

WHAT HAPPENS WHEN TRANSITION FAILS?

- **Failure** → non-adherence, ↑ emergency department visits, admissions, morbidity, mortality

TRANSITION AS A PROCESS

- **Six core elements**: policy 12–14 → tracking/readiness/ planning 14–18 → transfer 18–21 → complete 18–23
- **Readiness tool**: TRAQ (Transition Readiness Assessment Questionnaire)
- **Top preparation element**: how/where to find **valid information**
- **Private time alone** → sex, sexual function, sexual debut
- **Programmes**: Good 2 Go (Toronto), Ready Steady Go (UK), Got Transition (US)
- **Best programme** = the one that fits the unit
- **Age 26** = US limit on parent's insurance
- **No independent living** → plan medical custodianship

SYSTEMS AND MODELS OF CARE

- **Routes**: transition (1) + new adolescent problem (3) **ideal**; no-transition referral (2) suboptimal; **emergency after discharge** (4) **worst**

- **Lifelong paediatric urologist** → suits limited life expectancy; fails when patient outlives provider
- **Integrated paediatric + adult dept** → joint clinics; barriers: cost, geography, privileging
- **Direct transfer, no plan** → invisible until emergency
- **Multidisciplinary team (MDT) backbone**: nursing
- **Most complex MDT needs**: neurological disease (SB)

DEVELOPMENTAL CONSIDERATIONS

- **Genital surgery windows**: < 6 months (both); 2–3 y (theory) vs 4th–5th y (AAP 1975)
- **Stages**: bonding year 1; individuation 18–30 mo; phallic-oedipal 3–6 y
- **Augmentation cystoplasty**: child helps catheterise + wants to be dry
- **Exstrophy closure** (non-elective) → as soon as safe
- **Self-management components (4)**: adherence, responsibility, self-care skills, knowledge
- **Medical post-traumatic stress disorder**: no adult congenital urology data; cognitive-behavioural therapy mitigates (cardiac data)

THE PRACTICALITIES: SETTING UP A SERVICE

- **Training pathway**: none defined; usually paediatric + reconstructive
- **Core MDT**: paediatric + transitional urologist, nurses, coordinator, nephrologist, psychologist
- **Biggest adult-provider barrier**: unfamiliarity with the condition
- **Paediatric provider**: "Am I the best person?"

SERVICE LOGISTICS (CONT.)

- **Barriers 3–4**: institution (apathy, "not profitable") → infrastructure, philanthropy; health system (cost, inertia) → patients + marketing
- **Hub-and-spoke** ideal; telehealth limited by **dual state licensing**, no examination
- **Success metric proposed**: one attended post-transfer visit (AHRQ: no definition)

MALE GENITAL RECONSTRUCTION

- **Adult hypospadias repair complications**: 10.1–37.5 % first; 27.5–63.6 % redo
- **After paediatric repair**: cosmesis dissatisfied **17–36 %**; curvature ~12 %; ED ~19 %
- **Proximal**: ~half revised by puberty (median 16.5 y) → follow into adulthood
- **Recurrent chordee**: plication alone ~25 % vs ventral lengthening 5 %

- **BEEC:** dorsal chordee near-universal; full correction rare (diastasis, corporal disproportion)
- **Phalloplasty + prosthesis:** ~50 % complications
- **Hypospadias paternity** HR 0.79; semen impaired mainly with **undescended testis (UDT)** (↑ FSH)
- **BEEC male:** retrograde ejaculation ~half; > half **azoospermic**; 70.6 % of triers succeed, half with assisted reproduction (ART)
- **Regret drivers:** number of operations + final appearance, not meatal position; 54.7 % mild, 12 % moderate-strong
- **BEEC HRQoL** = general population; tracks **continence**

FEMALE BEEC / CLOACA / ANORECTAL MALFORMATION (ARM)

- **Cloaca:** void 46 %; incontinent 23 %; haematometra 38 %; primary amenorrhoea 25 %
- **Cloaca: chronic kidney disease (CKD) ~half; end-stage (ESRD) 17 %;** worst = **solitary kidney + reflux**
- **BEEC sex:** dyspareunia ½; FSFI dysfunction ⅓; **78 % satisfied**; cloaca 42 % sexually active
- **BEEC fertility:** 68 % conceive; **21 %** without ART in year 1
- **ARM:** Müllerian anomaly > ½; commonest **uterus didelphys**; ≥ ½ need ART
- **BEEC pregnancy:** loss ~20 %; fetal loss 5–7 %; UTI ⅓–⅔; **breech** (wide pelvis)
- **Reflux nephropathy pregnancy:** pregnancy-induced hypertension OR 5.55; pre-eclampsia OR 6.04; miscarriage 11–23 %
- **Pelvic organ prolapse 18–52 %;** acute postpartum in ½; pregnancy → pessary/bed rest

RECONSTRUCTION AND DIVERSION

- **Appendicovesicostomy** ~50 % fewer revisions than Monti; Mitrofanoff revision **30–50 %**
- **Stenosis commonest at skin;** fascia/valve → perforation
- **Can't catheterise:** overdilation kink → urethral/floppy-tip → needle via **old SP site**
- **Conduit stomal complications** 18–35 % (≤ 34.4 %); parastomal recurrence ≤ 56 % → **keyhole/Sugarbaker mesh**
- **Ureteroileal stricture ~10 %** → endoscopic first
- **BEEC after bladder neck reconstruction:** 59 % void; 64 % of those dry
- **Stones up to 40 %** (2 % with dependent drainage); < **1.5 cm** via channel; > **2 cm/multiple** open
- **10–12 Fr** catheters too small to irrigate; irrigation ≥ **240 mL**
- **Gastric segment** → **alkalosis**; ileum/colon → hyperchloraemic acidosis
- **Multiply operated:** 91.5 % complication risk

SURVEILLANCE AND UTI

- **SB CKD ¼–½;** only 62 % surveyed within 2 y
- **Highest CKD risk:** congenital obstructive nephropathy; PUV decline late adolescence
- **SBA:** no routine urodynamics after childhood; creatinine overestimates GFR if non-ambulant
- **Refer to nephrology:** eGFR ≤ **30**; loss ≥ **5/y**; ACR ≥ **300 mg/g**; only after urological optimisation
- **"Tempered" MAG3** = catheter in; not routine in adults

- **ISC bacteriuria 50–100 %;** treat symptoms only; > **4 UTIs/y** → investigate
- **Same organism** → source; **different** → reinoculation

ESRD AND TRANSPLANT

- **Congenital anomalies of kidney and urinary tract (CAKUT)** = 34–43 % paediatric ESRD; **posterior urethral valves (PUV)** 23 % ESRD by median 31
- **PUV:** ≤ ¼ decline post-puberty; ≤ 75 % bladder dysfunction
- **Polyuria mainstay:** overnight catheter; native nephrectomy discouraged
- **Native nephrectomy:** pyelonephritis, stones, malignant HTN, flank pain, nephrotic
- **Peritoneal dialysis relative contraindications:** adhesions, **ventriculoperitoneal shunt**
- **Transplant work-up:** cystoscopy + video urodynamics; willingness to self-catheterise
- **Post-transplant pyelonephritis** 0.49 vs 0.31–0.44/patient-year; **graft 92.3 % at 5 y**
- **BEEC:** avoid hypogastric artery → hemi-penile ED; **VACTERL** → MRA/CTA

CLASSIC TRAPS

- **Trap:** AHRQ shows transition futile — **Actually:** poor metrics; stability = success
- **Trap:** transfer = transition — **Actually:** transfer one step; complete 18–23
- **Trap:** all need lifelong follow-up — **Actually:** quiescent (VUR) may be discharged
- **Trap:** MDT need = lifelong need — **Actually:** exstrophy lifelong, urologist-led
- **Trap:** new adolescent problem suboptimal — **Actually:** route 3 ideal
- **Trap:** lifelong paediatric urologist best — **Actually:** fails as patient outlives provider
- **Trap:** elective timing follows theory — **Actually:** school, parental anxiety, surgeon
- **Trap:** weak processing → parent consents — **Actually:** capacity present; support consent
- **Trap:** confidentiality reviewed once — **Actually:** with family, then alone
- **Trap:** adult team starts transition — **Actually:** paediatric team initiates
- **Trap:** annual screening cystoscopy after cystoplasty — **Actually:** symptoms or filling defect only
- **Trap:** positive urine hCG after augment = pregnant — **Actually:** check serum hCG
- **Trap:** new continence is good news — **Actually:** hostile bladder; investigate
- **Trap:** treat all bacteriuria on ISC — **Actually:** symptomatic only
- **Trap:** creatinine reliable in wheelchair users — **Actually:** add cystatin C
- **Trap:** native nephrectomy for polyuria — **Actually:** overnight catheter first
- **Trap:** regret tracks severity — **Actually:** more with **distal**; driven by number of operations